

Pooja Kiran

Security Engineer | AI & Agent Security | Cloud IAM | Application Security
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EXPERIENCE

Independent AI Security Researcher & Engineer

Aug. 2024 - Present

Self-Directed Research

Tempe, AZ

- Started self-directed security research in Aug. 2024. A Dec. 2024 Flask/Docker OS resource-management lab later became a PostgreSQL-backed lease control plane; by 2026, the work expanded into agent runtime security, cloud IAM, model provenance, and training-data integrity across **19 repositories**.

MCP Agent Security Gateway | *Python, FastAPI, MCP/JSON-RPC 2.0, Elastic Security*

Jul. 2026 - Present

- Built an inline tool-call policy gateway with **55 prompt-injection patterns**, capability and PII/exfiltration checks, anti-SSRF, rate limiting, fail-closed enforcement, and tamper-evident audit evidence; validated **718 tests at 82.46% coverage**, **9 Elastic rules**, and **21 core SIEM tests**.

AWS Agent Identity Guard | *Python, AWS IAM, SARIF 2.1.0*

Aug. 2026 - Sep. 2026

- Built a static IAM analyzer with **25 deterministic rules** covering wildcards, `iam:PassRole`, `sts:AssumeRole`, privilege escalation, weak trust policies, audit tampering, and permission boundaries; verified **235 passing tests** with **3 credential-gated skips** and a synthetic **1.146 ms p95 / 1,846 policies/sec**.

HF Model Provenance Scanner | *Python, SafeTensors, GGUF, ONNX, Keras*

Jul. 2026 - Sep. 2026

- Built a non-executing model supply-chain scanner using pickle-opcode, AST/taint, dependency/provenance, obfuscation, and binary-format inspection; verified **241 tests at 75.81% coverage**, detected **33/33 committed adversarial fixtures**, and recorded **0 actionable findings across 4 committed benign samples**.

Dataset Poisoning Detector | *Python, FastAPI, Kafka, Redis, Prometheus*

Jul. 2026 - Present

- Built statistical and spectral training-data screening with Isolation Forest, label-aware analysis, drift detection, deduplication, streaming, and quarantine; verified **199 tests at 90.88% coverage** with a **90% CI gate** and **0.55 / 0.60 / 0.70 F1 at 5% / 10% / 20% poisoning**.

Business & Compliance Lead — AEROSEC

Aug. 2025 - Dec. 2025

Honeywell Aerospace Technologies × Arizona State University

Tempe, AZ

- Led business/compliance analysis for a proposed airline PSS protection layer and built a **\$120K first-year commercialization scenario** with a **five-year financial model** presented to ASU and Honeywell stakeholders.

Graduate Teaching Assistant — IT Grader

Jan. 2025 - Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University

Mesa, AZ

- Evaluated approximately **85 undergraduate web-development submissions per semester** and **52 graduate cybersecurity assignments** covering policy, compliance, risk analysis, and security controls.

EDUCATION

Arizona State University

Tempe, AZ

M.S., Information Technology (Security) – GPA: 3.87/4.00

Aug. 2024 - May 2026

M. S. Ramaiah University of Applied Sciences

Bengaluru, India

B.Tech., Computer Science & Engineering – CGPA: 8.44/10

Aug. 2019 - Aug. 2023

TECHNICAL SKILLS

Security: MCP/JSON-RPC, Tool Security, Prompt Injection, LLM Red Teaming, AWS IAM, API Security, Model Supply Chain, Training-Data Integrity, Adversarial ML

Engineering: Python, Rust, C++, FastAPI, pytest, Hypothesis, GitHub Actions, Docker, Kubernetes, SARIF 2.1.0, CodeQL, Bandit, Trivy

Detection: Elastic Security, ECS, SIEM, Prometheus, Security Telemetry, Tamper-Evident Audit Logging, MITRE ATLAS, OWASP LLM Top 10

Certifications: AWS Academy Cloud Architecting; AWS Academy Cloud Security Foundations